

Ready-to-Use Polymerization Simulations Combining Universal Machine Learning Interatomic Potential with Time-Dependent Bond Boosting for Polymer and Interface Design

Published as part of *The Journal of Physical Chemistry B* special issue "Physical Chemistry of Microplastics and Nanoplastics".

Hodaka Mori,* Shunsuke Tonogai, Yu Miyazaki, Akihide Hayashi, and Masayoshi Takayanagi



Cite This: *J. Phys. Chem. B* 2026, 130, 2710–2720



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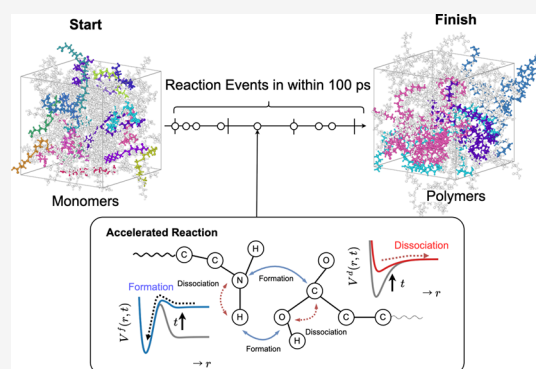


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ABSTRACT: Although polymerization and curing reactions govern the performance of advanced materials, their simulation remains challenging owing to the need for accurate, transferable potentials and the rarity of chemical events. Conventional reactive force fields such as ReaxFF require system-specific parametrization, while universal machine learning interatomic potentials (uMLIPs) exhibit limited sampling efficiency. This paper introduces a novel simulation framework integrating a uMLIP with a time-dependent bond-boost scheme. The bias potential increases monotonically with time, and the use of a unified parameter set across reaction classes enables consistent acceleration without system-specific tuning. For radical polymerization of vinyl monomers, the proposed framework reproduces characteristic trends, such as linear molecular-weight growth with conversion, initiator-concentration scaling, and relative monomer reactivity trends. For step-growth polycondensation of nylon-6,6, it captures the characteristic sharp increase in molecular weight at high conversion rates, consistent with experimental behavior. Epoxy curing on a copper substrate reveals interfacial ring-opening and cross-linking events, consistent with spectroscopic evidence of Cu–O–C bond formation. Overall, coupling uMLIPs with a time-dependent bond boost enables practical and transferable simulations of polymerization and curing processes. The proposed framework reliably resolves mechanistic pathways and relative reactivity, offering molecular-level insights into polymer growth and interfacial adhesion.



1. INTRODUCTION

Polymeric materials play an indispensable role across a wide range of industrial applications, including electronics, medical devices, energy storage, and structural components.^{1–4} Their performance strongly depends on the molecular structures formed by microscopic chemical processes such as polymerization and cross-linking, highlighting the importance of understanding structural changes and reaction mechanisms at the molecular level.^{5,6} Thus, to design high-performance materials and develop novel polymeric systems, the corresponding reaction pathways, intermediates, and transition states must be comprehensively explored.

Experimental techniques such as time-resolved spectroscopy and electron microscopy have been used to analyze polymer reactions.^{7,8} However, direct observation of reaction intermediates and detailed mechanisms at the atomic level remains challenging.^{9–12} Molecular simulations serve as powerful complementary tools that allow the identification of reaction intermediates and transition states that are difficult to capture experimentally.^{13–16} By providing atomistic insights, these

simulations offer valuable guidelines for the rational design of advanced materials.

Two methods are widely used for simulating cross-linking and polymerization reactions: pseudoreaction models and reactive potentials. Pseudoreaction models stochastically rearrange chemical bonds when specified distance or angular criteria are met.^{17–21} Although these models can be combined with classical molecular dynamics (MD) to handle large systems, they necessitate complex parameter tuning, including the definition of activation energies and frequency factors in advance. Quantum chemical approaches, such as global reaction route mapping, can accurately calculate activation

Received: December 2, 2025

Revised: January 19, 2026

Accepted: February 11, 2026

Published: February 18, 2026



energies. However, their computational cost remains prohibitive for large or structurally diverse condensed-phase systems.^{5,22,23}

A well-known reactive force field approach is ReaxFF, which describes the formation and breaking of chemical bonds during MD simulations.²⁴ However, many chemical reactions in polymerization processes are rare events, meaning that straightforward nanosecond-scale MD simulations often fail to capture sufficient reaction progress. This problem becomes particularly pronounced in polymerization, which involves a large number of sequential reactions, necessitating unrealistically long simulation times.²⁵ To address this, acceleration techniques such as bond-boost MD have been developed.^{26,27} However, the creation and parametrization of specialized force fields for each target system require substantial expertise and effort. Moreover, the bond-boost method involves hyperparameters that, if set improperly, may compromise the reliability of the observed reactions. Another widely used approach for enhancing the sampling of rare events is metadynamics, which applies history-dependent bias potentials to explore free-energy landscapes efficiently.^{28,29} However, by design, metadynamics promotes both forward and backward reactions, making it less suitable for directly accelerating one-way polymerization processes.

Given these limitations, there exists a strong demand for simulation methods that are simple, accurate, and broadly applicable to diverse polymerization processes. Recently, universal machine learning interatomic potentials (uMLIPs) trained on density functional theory data have attracted increasing attention in this domain.^{30–32} In particular, uMLIPs such as PreFerred Potential (PFP), M3GNet, CHGNet, and MACE have demonstrated remarkable generalization across various atomic species and structures,^{33–37} enabling their application to catalysis, metal–organic frameworks, and solid–liquid interfaces.^{38–41} The incorporation of these uMLIPs largely eliminates the need to construct system-specific force fields, thereby broadening the scope of atomistic simulations.

Considering these aspects, this article introduces a time-dependent reaction acceleration strategy that combines the broad transferability of a uMLIP with efficient sampling. This approach enables polymerization simulations that are both accurate and straightforward to implement, even for chemically complex systems. To validate the method, we first benchmark the radical polymerization of vinyl monomers, focusing on the kinetics and relative reactivity. We then demonstrate its broad applicability through step-growth polycondensation of nylon-6,6 and the curing of an epoxy resin at a copper oxide (CuO) interface, a chemically heterogeneous setting in which reactive force fields (e.g., ReaxFF) often require careful selection and validation and, where necessary, refinement of the force-field parameter set to achieve a balanced description of organic curing chemistry and oxide surface reactivity. Recently, ReaxFF parameter sets for Cu/C/H/O surface chemistries and epoxy–amine curing chemistries have been reported, and accelerated ReaxFF studies have demonstrated that bond-boost concepts can be coupled with reactive MD to access otherwise rare events.^{15,27,26–43} Nevertheless, because such parameter sets are typically optimized for broad materials and reaction classes, ensuring a consistently reliable description of epoxy–amine curing, hydroxylated CuO surface reactions, and interfacial bond formation under the specific conditions considered here may require substantial system-specific validation and, in some cases, additional refinement. In this context, transferable

uMLIPs—trained on diverse first-principles data and capable of representing a wide range of chemical environments with a single model—offer an attractive alternative for chemically heterogeneous polymer–interface systems, where expressive power and generality are critical.

2. COMPUTATIONAL DETAILS

2.1. Time-Dependent Bond-Boost Potential

To accelerate reactions, we use a modified version of the bond-boost method originally proposed by Vashisth et al.²⁶ The original bond-boost potential is expressed as

$$V(r) = f_1 \{1 - \exp[-f_2(r - r_0)^2]\} \quad (1)$$

Here, f_1 , f_2 , r_0 , and r represent the depth of the boost potential, its range, the target distance, and the interatomic distance, respectively. Appropriate values for f_1 , f_2 , and r_0 must be specified prior to the simulation. Vashisth et al.²⁶ and Dasgupta et al.²⁷ examined multiple parameter sets and adopted the acceleration conditions that best reproduced the activation energies obtained using ReaxFF. Because bond-boost acceleration relies on the underlying potential to approximately preserve the relative ordering of reaction barriers (and thus pathway selectivity), the fidelity of barrier descriptions is crucial. While ReaxFF parameter sets are often parametrized and validated against reactive training data, the transition-state coverage of universal MLIPs depends on the training data set; therefore, barrier fidelity should be assessed on a case-by-case basis. Therefore, we provide an initial assessment of barrier fidelity for representative propagation reactions via explicit transition-state searches with PFP (Table S4). Typically, determining f_1 requires estimation of the activation energy in advance and then fine-tuning of the parameters through trial and error. Furthermore, for bond cleavage, the appropriate distance r_0 must be defined as the threshold for bond breaking. As the number of target reactions increases, such preliminary parameter studies become increasingly cumbersome, complicating the overall simulation setup.

To address this, in this work, we introduce a time-dependent bond-boost (TDBB) method as a modification of the original approach. The key innovation is that the bias potential increases with time, as shown in eqs 2–5. Here, $V^f(r, t)$ and $V^d(r, t)$ represent the bias potentials applied to promote **bond formation** and **bond dissociation**, respectively:

$$V^f(r, t) = f_1(t) \{1 - \exp[-f_2(r - r_0)^2]\} \quad (2)$$

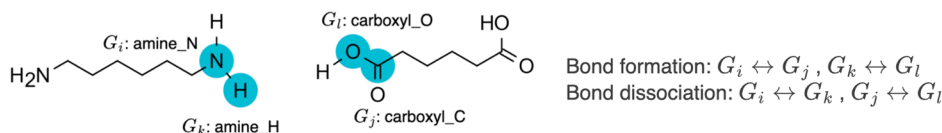
$$V^d(r, t) = f_1(t) \exp(-f_2 r^2) \quad (3)$$

$$r_0 = \lambda \sum_{a \in \{\text{pair}\}} r_a^{\text{vdw}} \quad (4)$$

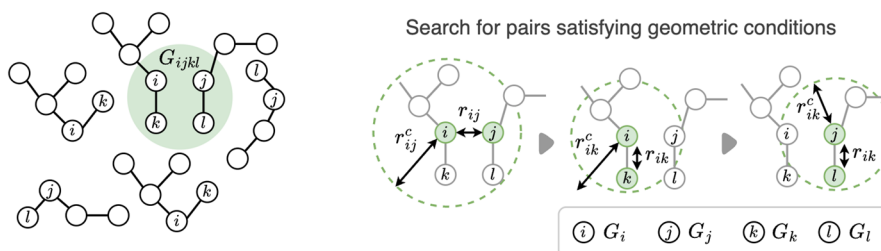
$$f_1(t) = \begin{cases} \gamma t, & f_1(t) < f_1^{\text{max}}, \\ f_1^{\text{max}}, & f_1(t) \geq f_1^{\text{max}} \end{cases} \quad (5)$$

where γ is the boost factor, r_a^{vdw} is the van der Waals radius of atom a (summed over the two atoms forming the reactive pair), λ is a scale factor defining the bonding threshold, and f_1^{max} is the maximum boost amplitude. Because $f_1(t)$ increases with time, reactions are gradually promoted as the simulation proceeds. Moreover, applying a single set of boost parameters to all possible reactions ensures consistent acceleration and the

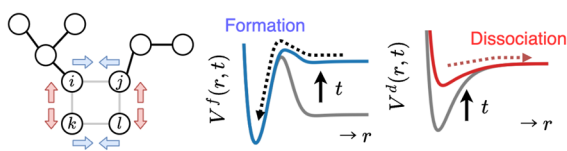
1. Define reactive groups



2. Search acceleration targets



3. Induce reaction by running boosted MD with NNP



4. Relax and check reaction

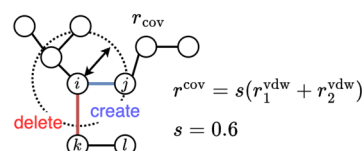


Figure 1. Schematic of the proposed simulation workflow, including four main stages: (1) definition of reactive atomic groups, where potential reactive species are identified from chemical functionalities; (2) selection and acceleration of reaction-targeted groups, where candidate atomic combinations satisfying geometric and chemical criteria are chosen and subjected to time-dependent bias potentials; (3) reaction induction and structural relaxation, in which accelerated molecular dynamics simulations promote bond rearrangements followed by unbiased relaxation; and (4) reaction evaluation and group update, where the bonding states are analyzed and reactive groups are updated for subsequent iterations.

capturing of relative reactivities without system-specific parametrization. The overall acceleration level can be tuned by γ , with larger values yielding higher reaction rates.

The rationale for a monotonic, time-growing bias is as follows: In conventional bond-boost MD, the bias depth is fixed throughout an acceleration window and must be comparable to (but not excessively larger than) the activation barrier of the targeted reaction. In practice, this often requires either prior barrier estimates or iterative trial-and-error tuning, particularly when multiple reaction types with distinct barriers coexist. The time-dependent form in eq 5 can be interpreted as an automatic, on-the-fly scan of the boost amplitude: The dynamics are only weakly perturbed at early times, and the bias increases gradually only if the system fails to react, eventually triggering an event without per-reaction tuning. This design improves usability in chemically heterogeneous systems and enables a simple default setup, where a single parameter set can be applied across reaction classes. Furthermore, because the bias can eventually exceed the barrier, this strategy trades physical time fidelity for sampling efficiency; therefore, trajectories should be interpreted primarily in terms of mechanistic pathways and relative trends rather than absolute rate constants.

Notably, we employ distinct functional forms for bond formation, $V^f(r_{ij})$, and bond breaking, $V^d(r_{ij})$, to ensure that once the product or dissociated state is reached, the bias potential vanishes and normal MD resumes. In the bond-formation case, the potential drives two atoms to approximately 60% of the sum of their van der Waals radii. Once that distance is reached, the potential effectively becomes zero. In the bond-breaking case, once the bond extends beyond a

critical length, the boost potential similarly disappears. This design eliminates the need to define separate parameters for each reaction, thereby simplifying the simulation setup.

According to eqs 2–5, the parameters f_1 , f_2 , and r_0 correspond closely to those in the original bond-boost method. Although these parameters can be fine-tuned if desired, $f_1^{\max}$ primarily serves to cap the boost strength and prevent excessively large acceleration. Similarly, f_2 controls how the boost varies with distance: Smaller values broaden the spatial extent of the bias, whereas larger values confine the bias to shorter ranges, with the characteristic width scaling as $1/\sqrt{f_2}$. By setting $r_{0,\text{create}}$ to the sum of the van der Waals radii of the involved atoms, we determine this parameter as independent of the reaction pathway. In practice, the same settings have been successfully applied to multiple reactions.

The basic functional form follows the original bond-boost design, ensuring that once a reaction occurs, the boost potential decays to zero and the system behaves as in a standard MD simulation without acceleration. An additional advantage of this straightforward Morse-like potential is that it consistently exerts force along the direction of the intended reaction coordinate. In contrast, adaptive bias methods, such as metadynamics, can, in some cases, drive the system in the opposite direction of the desired pathway. Although adding a high-wall restraint can prevent the bias from diverging, such methods typically fill energy minima before effectively accelerating the system toward the reaction coordinate, potentially delaying the onset of meaningful acceleration. Moreover, in metadynamics, the bias can continue to accumulate even after a reaction has occurred, necessitating

Table 1. Comparison of Acceleration Methods, Focusing on Bias Form and Setup Requirements

method	setup requirements	bias function	reaction directionality	time rescaling
bond boost ²⁶	reactive groups, bias height, target distance	fixed bias	unidirectional	no
metadynamics	collective variables (CV)	adaptive hill bias	bidirectional	no ^b
CVHD	CV and dividing surface	adaptive hill bias	bidirectional	yes ^a
TDBB (this work)	reactive groups	monotonic, time-growing bias	unidirectional	no

^aHyperdynamics time rescaling is possible if conditions are met (zero bias at dividing surface, bias < barrier, TST assumptions). ^bExcept infrequent metadynamics.

additional steps (e.g., detection of the occurrence of the reaction) to halt bias deposition.

Lastly, the proposed method does not allow time rescaling, unlike conventional hyperdynamics.^{44–46} According to transition-state theory, time rescaling in biased MD is valid only if (1) the added potential does not exceed the activation energy and (2) no potential is applied at the dividing surface. In contrast, in the proposed approach, the bias grows monotonically with time and can surpass the activation barrier regardless of the potential energy landscape. Thus, these conditions are generally not satisfied. Notably, while this method simplifies parameter setup and effectively drives chemical reactions, the time evolution of the trajectory no longer strictly corresponds to physical time once the bias becomes substantial.

2.2. Polymerization Procedure

The proposed simulation method involves four main steps (Figure 1): (1) definition of reactive atomic groups, (2) selection and acceleration of reaction-targeted groups, (3) reaction induction and structural relaxation, and (4) evaluation of reactions and updating of atomic groups.

2.2.1. Definition of Reactive Atomic Groups. In the initial stage, reactive atomic groups are identified and defined as sets G_I , G_J , G_K , and G_L , each corresponding to a specific group (e.g., oxygen atoms in epoxy groups, nitrogen atoms in amine groups, or α -carbon atoms in vinyl groups for radical polymerization). These sets are defined as

$$G_X = \{a|a \in X\}, \quad X = I, J, K, L \quad (6)$$

with a representing an individual atom belonging to the group. Chemical reactions are modeled as rearrangements of atomic bonds represented by bond formation and breakage.

2.2.2. Selection and Acceleration of Reaction-Targeted Groups. To efficiently identify possible reaction candidates, a two-stage selection process is employed. First, neighboring atomic groups are preselected based on geometric proximity to identify potential reacting partners. Subsequently, based on reaction rules, candidate groups are formed by selecting tuples (i, j, k, l) such that $i \in G_I$, $j \in G_J$, $k \in G_K$, and $l \in G_L$, with interatomic distances constrained to the window $r_{ab}^{\min} \leq r_{ab} \leq r_{ab}^{\max}$. Defining the set of target pairs as $\mathcal{P} = \{(i, j), (i, k), (j, l)\}$, the candidate set is expressed as

$$G_{ijkl} = \{(i, j, k, l)|i \in G_I, j \in G_J, k \in G_K, l \in G_L, r_{ab} \in [r_{ab}^{\min}, r_{ab}^{\max}] \quad \forall (a, b) \in \mathcal{P}\} \quad (7)$$

Moreover, we allowed multiple reaction types to coexist. For each reaction type r , a candidate set $G_{ijkl}^{(r)}$ is prepared with its own group assignments (G_I , G_J , G_K , G_L), target-pair set $\mathcal{P}^{(r)}$, and distance windows $[r_{ab}^{\min}, r_{ab}^{\max}]^{(r)}$. This setup enables systems in which different reactions proceed simultaneously within a single simulation box (e.g., dehydration-condensation and radical polymerization occurring in parallel). Unless otherwise

noted, candidates from all reaction types are pooled and subjected to the same nonoverlapping selection and biasing procedure described below.

As candidate groups may share atoms (e.g., (0, 1, 2, 3) and (0, 4, 5, 6)), a nonoverlapping selection is performed by computing the score $d_{ijkl} = r_{ij} + r_{ik} + r_{jl}$ for each group, sorting the obtained scores in ascending order, and sequentially selecting groups while skipping those with overlapping atoms. This yields the refined set G_{ijkl}^{acc} . For each group in G_{ijkl}^{acc} , the set of reactive atomic pairs P is then defined, consisting of one or more pairs selected from (i, j) , (i, k) , (j, l) , and (k, l) that participate in bond formation or cleavage. A time-dependent bias potential is subsequently applied to each pair in P :

$$\Delta V(t) = \sum_{(i,j,k,l) \in G_{ijkl}^{\text{acc}}} \sum_{p \in P} [f_p V^f(r_p, t) + (1 - f_p) V^d(r_p, t)] \quad (8)$$

where $f_p = 1$ denotes bond formation, $f_p = 0$ denotes bond breakage, and r_p is the interatomic distance of pair p .

2.2.3. Reaction Induction and Structural Relaxation. The system is subjected to an accelerated MD simulation under the applied bias $\Delta V(t)$ for either a predetermined period or until a reaction event is observed. A reaction event is defined to occur when the interatomic distances of a specific set of pairs (ij , ik , and jl) simultaneously satisfy the prescribed bonding conditions. For each pair, the bonding state is determined based on the van der Waals criterion: Two atoms are considered bonded if their separation falls below 60% of the sum of their van der Waals radii. If $f_p = 1$, the event is triggered when these pairs newly satisfy the bonding condition (bond formation). If $f_p = 0$, the event is triggered when the pairs cease to satisfy the condition (bond dissociation). Once such an event is detected or the maximum simulation time is reached, the acceleration phase terminates. Subsequently, a standard MD simulation (without bias) is performed to allow structural relaxation and equilibration of the resulting configurations.

2.2.4. Reaction Evaluation and Group Update. After structural relaxation, the system is analyzed to compare bonding states before and after the biased simulation. Atoms that have participated in new bond formations or bond cleavages are marked as “reacted” and removed from the reactive set. In contrast, atoms that experienced the bias but did not undergo any bond changes remain in the reactive set for subsequent iterations. If the resulting species remain reactive (e.g., in epoxyamine curing or vinyl radical polymerization), these atoms are reassigned to Group X and considered in subsequent iterations. Repeating these steps iteratively drives the overall polymerization process.

2.3. Comparison with Existing Acceleration Methods

To highlight the novelty and advantages of the proposed framework, Table 1 summarizes the key features of the widely

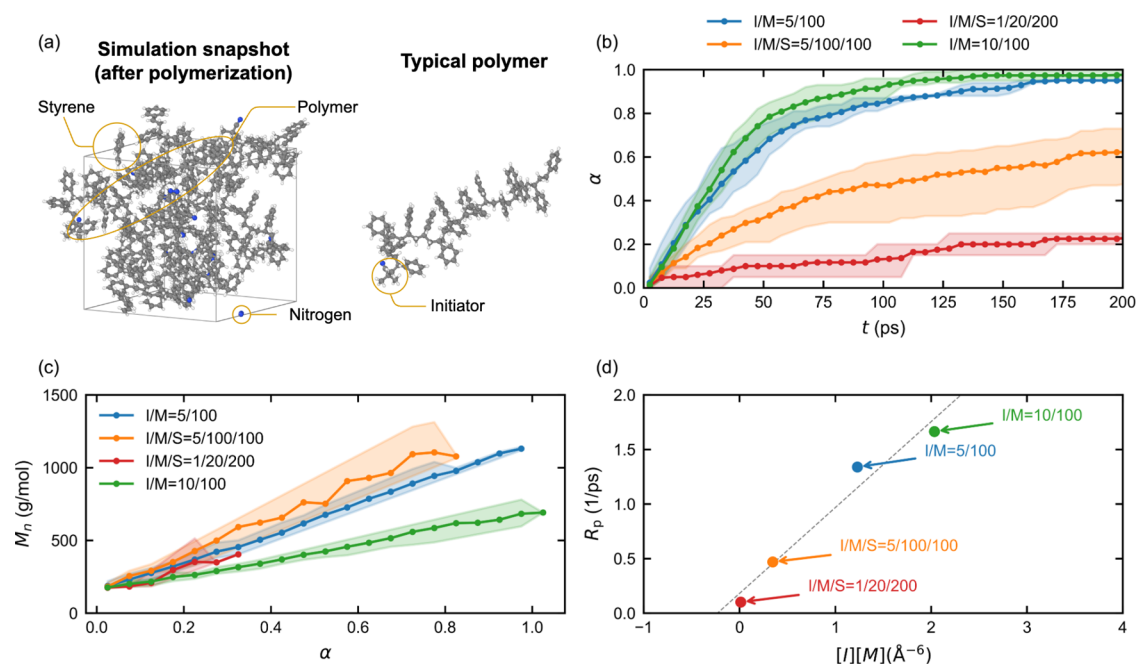


Figure 2. Structural and kinetic characteristics of styrene radical polymerization in toluene solvent. (a) Snapshot after polymerization and a representative polymer chain. (b) Time-conversion curves at different initiator-to-monomer (I/M) ratios. The vertical axis α represents the monomer conversion, defined as $\alpha = 1 - [M]/[M]_0$, where $[M]_0$ and $[M]$ are the initial and instantaneous monomer concentrations, respectively. (c) M_n as a function of conversion. (d) Polymerization rate plotted against $[I][M]$. The horizontal axis represents the product of the initiator and monomer concentrations, $[I][M]$, converted from number densities to molar units ($1 \text{ \AA}^{-6} = 2.76 \times 10^{-4} \text{ mol}^2 \text{ L}^{-2}$). Here, I , M , and S denote the initiator (AIBN), monomer (styrene), and solvent (toluene), respectively.

used rare-event acceleration techniques. Conventional bond boost requires system-specific parameter tuning based on activation barriers, whereas hyperdynamics and metadynamics depend on carefully chosen collective variables and typically accelerate both forward and backward events indiscriminately. In particular, even infrequent metadynamics—the only variant that, in principle, allows approximate recovery of physical time scales—is intrinsically bidirectional: Gaussian hills are deposited regardless of whether the system proceeds forward or backward, which makes the method inefficient and physically inconsistent for inherently unidirectional processes such as radical polymerization. In contrast, bond-boost schemes apply a forward-only bias that selectively promotes bond formation, aligning with the irreversible nature of polymer growth. Our TDBB variant retains this property while eliminating the need for prior parametrization, allowing a single parameter set to be used across reaction types. Combined with uMLIP such as PFP, this approach offers a unique balance of simplicity, accuracy, and transferability, and is particularly well-suited for chemically complex and heterogeneous polymerization systems.

3. METHODS

We investigated three representative systems: radical polymerization of vinyl monomers, step-growth polycondensation of nylon-6,6, and epoxy curing at the CuO interface. All systems were prepared through a two-stage equilibration procedure: initial relaxation with a classical force field, followed by refinement using the PFP uMLIP under the NPT ensemble. Detailed compositions and equilibration protocols are provided in the Supporting Information.

All simulations were conducted using PFP (version 6.0) in the CRYSTAL_PLUS_D3 mode. PFP is a uMLIP trained on Perdew–Burke–Ernzerhof reference data and augmented with the D3 dispersion correction, which improves the reproducibility of

condensed-phase properties such as density. Owing to its transferability across 96 elements, PFP has been applied to a wide range of materials, including inorganic and organic compounds, polymers, and catalytic systems.

Reactive acceleration MD was performed using the boost potential described in Section 2.1. Unless otherwise noted, simulations employed a time step of 0.25 fs, alternating biased and unbiased dynamics every 2000 steps (500 fs), and bias parameters of $f_1^{\text{max}} = 250 \text{ kcal/mol}$ for V^d , $f_1^{\text{max}} = 125 \text{ kcal/mol}$ for V^d , $f_2 = 10 \text{ \AA}^{-2}$, and $\gamma = 1.0$. Reaction events were identified based on van der Waals radii, as detailed in Section 2.1. For each condition, three independent simulations with randomized initial configurations were performed, and the reported results represent averages over these runs. System-specific simulation parameters, including system size, composition, and equilibration protocol, are provided in the Supporting Information.

To evaluate the robustness of the acceleration scheme, we examined the sensitivity of the cumulative number of reactions to variations in f_1^{max} , f_2 , and γ , using styrene radical polymerization as a representative system (Figure S4). The influence of f_1^{max} was negligible once it exceeded a moderate threshold: extremely low values suppressed reactions within the time window, whereas sufficiently large values simply shifted the reaction to occur later within the same window, after which the reactive-pair list was updated. System behavior was also insensitive to variations in the f_2 . Although f_2 formally defines the spatial range of the bias, propagation is governed by near-contact events, and $f_2 \approx 10$ was sufficient to ensure consistent behavior. In contrast, increasing γ accelerated all of the reaction channels uniformly, acting as a global rate scaling factor. These analyses demonstrate that the default parameter set is broadly representative and does not distort relative reactivity trends.

All MD simulations were carried out using OpenMM ver. 8.1.1.⁴⁷ Forces and energies were evaluated using PFP, with GPU selection depending on the system size: an NVIDIA Tesla V100 (16 GB memory) was used for medium-sized systems, whereas larger systems were computed on an NVIDIA A100 (80 GB memory). A custom interface coupled OpenMM with PFP. The bias potential was

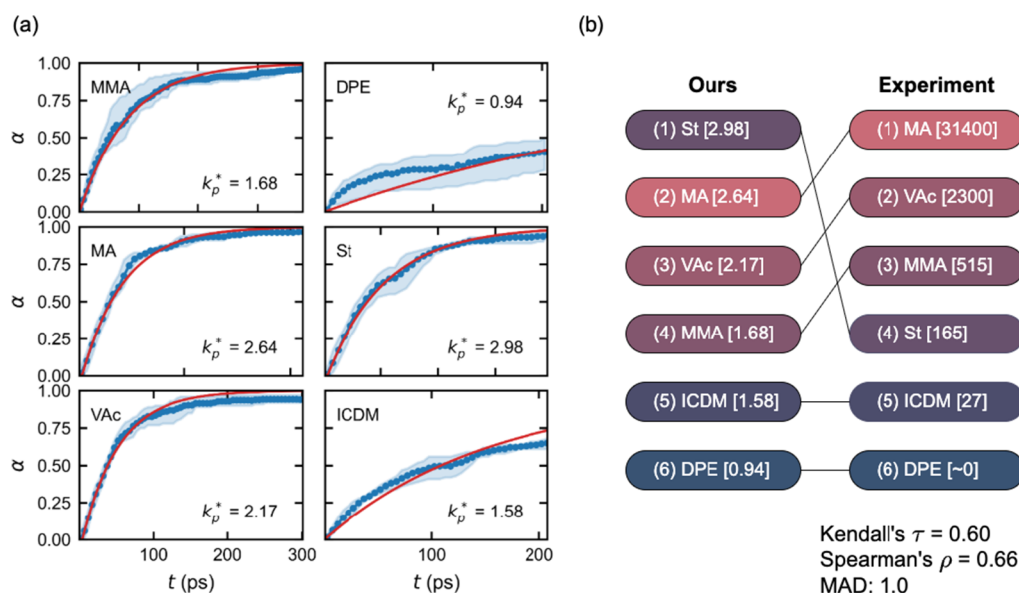


Figure 3. (a) Time–conversion profiles (blue) and exponential fits (red) for representative vinyl monomers, with extracted apparent propagation rate constants k_p . (b) Ranking of the apparent k_p values compared to experimental data. Rank correlations indicate Kendall's $\tau = 0.60$, Spearman's $\rho = 0.66$, and mean absolute deviation (MAD) = 1.0 in rank position (see definition in text).

implemented via OpenMM–Torch, enabling TorchScript-defined energy and force terms to be directly incorporated. Force derivatives were evaluated using PyTorch's automatic differentiation, eliminating the need for manual implementation. The workflow of reactive acceleration MD was managed by using a Python driver script. For reproducibility, the workflow logic (including candidate generation, nonoverlapping selection, event detection, and group updates) and key parameters are summarized within a pseudocode in the Supporting Information (Algorithm S1).

4. RESULTS AND DISCUSSION

4.1. Radical Polymerization of Vinyl Monomers

The simulations were designed with two primary objectives: (i) capture qualitative kinetic trends of radical polymerization under solution conditions (e.g., dependence of apparent reaction progress on initiator/monomer concentrations) and (ii) reproduce the relative reactivity order of representative vinyl monomers under bulk conditions. The monomer set included methyl acrylate, methyl methacrylate, styrene, vinyl acetate, diphenylethylene, and dimethyl itaconate. Polymerizations were performed both in toluene solution and in the bulk phase. Toluene was selected as a representative nonpolar solvent commonly used in experimental radical polymerizations, whereas bulk simulations enabled direct assessment of the solvent-free reactivity. Together, these two environments and the diverse monomer set provided a stringent benchmark for evaluating the predictive capability of the proposed simulation framework.

Figure 2 illustrates three key aspects of the styrene radical polymerization under varying concentrations: (i) time–conversion profiles, (ii) relationship between monomer conversion and number-average molecular weight (M_n), and (iii) dependence of the polymerization rate on the initiator-to-monomer ratio (I/M). As shown in Figure 2b, systems with I/M ratios of 5/100 and 10/100 exhibit markedly higher conversions at earlier times compared to those with lower initiator loadings. This trend demonstrates that increasing the initiator and/or monomer concentrations accelerates polymerization, consistent with classical kinetic expectations and

experimentally reported concentration dependences in free-radical polymerization. These results highlight the ability of the proposed uMLIP–accelerated MD framework to qualitatively capture early-stage kinetic trends of radical polymerization.^{8,48}

The plots of conversion versus M_n in Figure 2c exhibit a clear linear relationship across all simulated conditions. This linear trend strongly suggests that, in the absence of termination and chain-transfer reactions (Figure S1), polymerization proceeds in a “living” manner, characterized by continuous chain growth without significant loss of active chain ends. Moreover, the slope of the conversion–molecular-weight curve for the $I/M = 10/100$ system is approximately half of that for the lower-initiator systems. This behavior is characteristic of living polymerization, where the number-average degree of polymerization is inversely proportional to the initiator concentration. This finding corroborates the fidelity of the proposed simulations in reproducing mechanistic features observed experimentally in controlled polymerization systems.

Figure 2d illustrates the dependence of the polymerization rate on the I/M ratio. In the absence of termination reactions, the polymerization rate R_p is expressed as

$$R_p = -\frac{d[M]}{dt} = k_p[M][P^*] \quad (9)$$

$$R_p \propto [M][I] \quad (10)$$

where $[M]$ and $[I]$ denote the monomer and initiator concentrations, respectively; $[P^*]$ is the concentration of propagating radicals; and k_p is the propagation rate constant. In this idealized case, all radicals generated from the initiator remain active, resulting in a linear dependence of the polymerization rate on the initiator concentration. This is unlike the square root dependence observed in conventional radical polymerizations.

The simulation results closely follow this theoretical proportionality, confirming that the proposed approach reproduces the linear dependence of the polymerization rate on initiator concentration and its proportionality to monomer

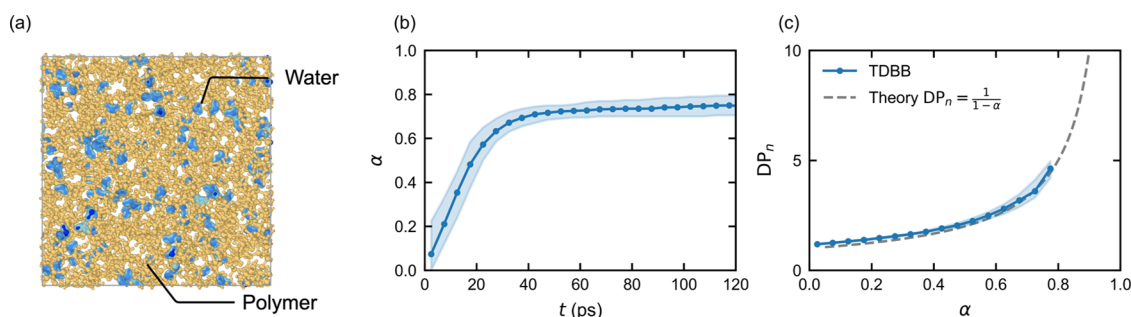


Figure 4. Structural and kinetic characteristics of nylon-6,6 step-growth polymerization under melt conditions without water removal. (a) Representative snapshot after polymerization, showing polymer chains (tan) and generated water molecules (blue). (b) Time evolution of monomer conversion (α), which levels off below unity owing to equilibrium limitation. (c) Number-average degree of polymerization (DP_n) as a function of conversion compared with the theoretical Carothers relation. The simulations reproduce the characteristic nonlinear DP_n –conversion relationship and incomplete conversion observed experimentally in condensation polymerizations without continuous removal of the condensate.

concentration under the no-termination assumption. This agreement between simulation and theory supports the idea that the proposed framework captures the expected scaling behavior under the present no-termination assumption, underscoring the robustness of the acceleration scheme in reproducing fundamental qualitative kinetic trends across a range of concentrations.

Collectively, these results demonstrate that combining machine-learned potentials with accelerated dynamics can reproduce key mechanistic features and the expected kinetic scaling trends in radical polymerization. By capturing the expected dependence of reaction rate and molecular weight evolution on initiator and monomer concentrations, the proposed framework provides a useful basis for trend-level/comparative simulations of polymerization and offers valuable insights into the molecular-level dynamics governing polymer growth.

Figure 3 presents the apparent propagation rate constants (k_p^*) for various vinyl monomers. The apparent rate constants were obtained by fitting the monomer conversion profiles to a single-exponential function,

$$\alpha(t) = 1 - \exp(-k_p^* t) \quad (11)$$

where $\alpha(t)$ is the monomer conversion at time t , and k_p^* represents the effective propagation rate under the simulated conditions. Representative fits are shown in Figure 3a, together with the extracted k_p values for each monomer. These fitted values were then ranked and compared against experimental propagation rate constants (Figure 3b). Rank correlations between the simulated and experimentally reported propagation rates yield a Spearman correlation coefficient (ρ) of 0.66, a Kendall rank correlation (τ) of 0.60, and a mean absolute deviation (MAD) of 1.0 in rank position, defined as $MAD = \frac{1}{N} \sum_i |\text{rank}_{\text{sim}}(i) - \text{rank}_{\text{exp}}(i)|$. Here, $\text{rank}_{\text{sim}}(i)$ and $\text{rank}_{\text{exp}}(i)$ denote the ranks of monomer i based on the simulated and experimental propagation rate constants, respectively; and N is the number of distinct monomers considered.

Because bond-boost acceleration assumes that the underlying potential approximately preserves the relative ordering of reaction barriers (and thus pathway selectivity), we provide an initial assessment of barrier fidelity for representative propagation reactions via explicit transition-state searches with PFP (Table S4). The predicted barrier ordering does not fully match the experimental trends. For example, PFP

ranks styrene as having the lowest propagation barrier among the studied monomers, whereas experimental estimates place styrene closer to the middle of the series. Such differences in barrier ordering can contribute to deviations in the simulated propagation rate constants. However, the correspondence between barrier errors and deviations in the apparent propagation rate constants is not one-to-one. In condensed-phase polymerization, propagation kinetics are influenced not only by activation barriers but also by configurational, entropic, and diffusional factors. Although bulk densities are reproduced reasonably well (Table S4), their ranking is not perfectly consistent with experiments, indicating that condensed-phase structural properties alone do not fully account for the observed deviations. Moreover, because the applied time-dependent bias alters the effective time scale of the dynamics, the present approach is not intended to reproduce absolute propagation rate constants. Instead, its primary utility lies in capturing relative reactivity trends and qualitative rankings across monomers without requiring prior knowledge of the activation energies. Improving barrier fidelity in uMLIPs, for example, through training data sets that more explicitly include transition-state information, is therefore expected to improve—but not solely determine—the quantitative agreement of kinetic rankings.

Nonetheless, rank inversions are observed for certain monomers, highlighting areas where the proposed model struggles to capture subtle differences in propagation kinetics. The range of propagation rate constants examined corresponds to activation energy variations of only a few kcal/mol. Accurately resolving such fine energetic differences will likely require higher-fidelity uMLIP trained on more comprehensive and accurate reference data or advanced acceleration schemes capable of capturing the delicate energetics of reactive events. Notably, because the applied bias alters the effective time scale of the dynamics, the proposed approach is not intended to reproduce absolute propagation rate constants. Rather, its primary utility lies in capturing relative trends and reactivity rankings across monomers without requiring prior knowledge of the activation energies. Incorporating higher-fidelity potentials or refined acceleration strategies is expected to further improve quantitative agreement and expand the applicability of this framework to complex polymerization systems.

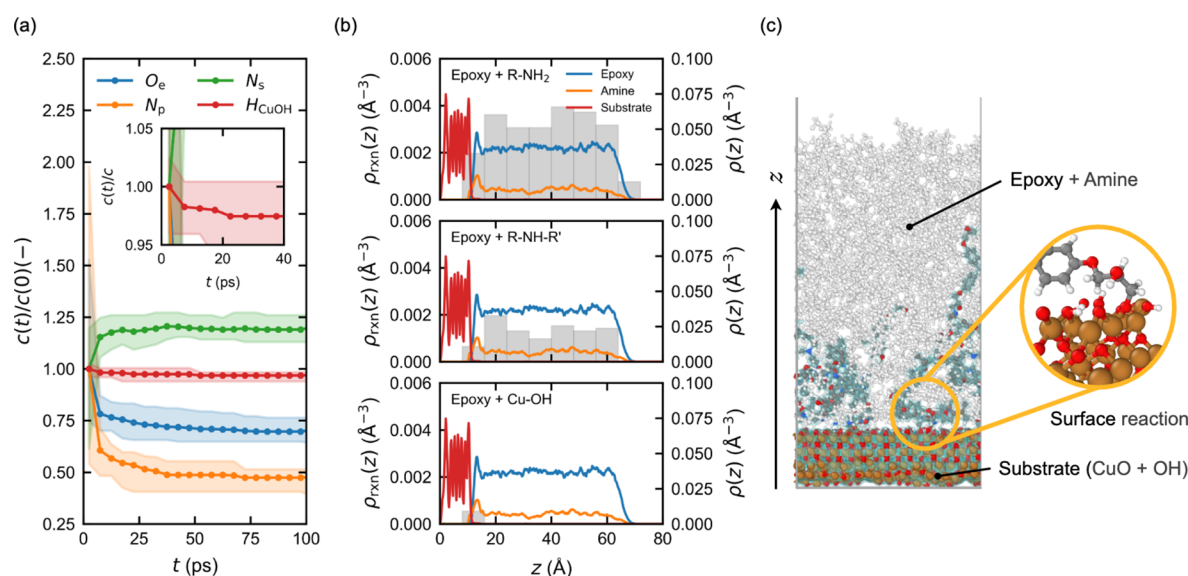


Figure 5. Epoxy curing at the hydroxylated CuO interface. (a) Time evolution of relative concentrations $c(t)/c(0)$ of epoxy oxygen (O_e), primary amine nitrogen (N_p), secondary amine nitrogen (N_s), and surface hydroxyl hydrogen (H_{CuOH}). (b) Depth-resolved reaction density $\rho_{rxn}(z)$ for epoxy–primary amine (top), epoxy–secondary amine (middle), and epoxy–surface reactions (bottom), together with atomic number density profiles $\rho(z)$ of epoxy (blue), amine (orange), and CuO substrate (red). (c) Representative snapshot of the epoxy/CuO interface with magnified view showing covalent bond formation between epoxy and surface hydroxyl groups.

4.2. Step-Growth Polycondensation of Nylon-6,6

To further evaluate the capability of the proposed framework in modeling step-growth polymerization, we examined the polymerization of nylon-6,6, a prototypical condensation polymer. The results (Figure 4) show the time evolution of monomer conversion and the relationship between conversion and DP_n . Here, DP_n denotes the number-average degree of polymerization defined as the average number of monomeric units per polymer chain. DP_n is related to the number-average molar mass M_n by $DP_n = M_n/M_0$, where M_0 is the molar mass of the repeating unit. Consistent with classical theories of step-growth polymerization, DP_n increases monotonically with conversion, particularly in the later stages, dominated by oligomer–oligomer coupling. This trend, a hallmark feature of step-growth kinetics, is reproduced with high fidelity across all runs, underscoring the robustness of the methodology. The quantitative agreement between simulation and theoretical expectations indicates that the uMLIP–accelerated framework accurately captures the essential thermodynamic and kinetic characteristics of condensation polymerizations.

Our results also align with experimental observations of nylon polycondensation. In melt polymerization without continuous water removal, conversion typically levels off at 60–80%.⁴⁸ Our simulations reproduce this incomplete conversion and limited chain length, in contrast to idealized models that assume the perfect removal of the condensate. This correspondence highlights the ability of the proposed framework to capture the equilibrium-limited nature of step-growth polymerizations without requiring predefined activation barriers or frequency factors for each reaction pathway, demonstrating a clear advantage over heuristic methods.

4.3. Epoxy Curing at the CuO Interface

Reliable adhesion between epoxy resins and metallic wiring layers is essential for improving the performance and reliability of semiconductor devices. Surface treatments such as oxygen plasma are commonly applied to enhance adhesion, and

spectroscopic studies have revealed the presence of Cu–O–C linkages at treated interfaces.⁴⁹ These observations suggest that covalent bond formation across the epoxy–copper interface is a key factor underlying the improved interfacial strength. However, molecular-level understanding of these interfacial curing reactions remains challenging to achieve owing to their chemical complexity and large system sizes.

This section describes the assessment of the ability of the proposed accelerated simulation framework to capture interfacial curing chemistry. The considered case is particularly demanding because it requires a potential that simultaneously captures epoxy–amine curing reactions, hydroxylated CuO surface chemistry, and interfacial bond formation in a large heterogeneous system. While ReaxFF parametrizations for Cu/C/H/O/N systems have been reported, and accelerated ReaxFF studies have demonstrated bond-boost concepts for accessing rare events,^{15,27,26–43} extending and validating a single parameter set for the present hydroxylated CuO/epoxy–amine interface under the conditions considered here remains nontrivial. Full ab initio dynamics at this scale are computationally infeasible. Therefore, we adopt a transferable uMLIP (PFP) as a practical alternative; however, straightforward uMLIP MD still suffers from a rare-event bottleneck, failing to capture curing reactions within accessible time windows. To analyze the curing reaction, Figure 5a shows the relative concentration change $c(t)/c(0)$ of key atomic species as a function of simulation time:

$$c(t)/c(0) = \frac{N_{\text{species}}(t)}{N_{\text{species}}(0)}$$

Here, $N_{\text{species}}(t)$ is the number of target atoms at time t . Both epoxy oxygen (O_e) and primary amine nitrogen (N_p) decrease over time, while secondary amine nitrogen (N_s) increases. This indicates that epoxy ring-opening by primary amines is the dominant reaction in the early stage. The increase in N_s confirms the formation of secondary amines, consistent with

the expected initial step of epoxy curing. Experimental studies of diglycidyl ether of bisphenol A/diethylenetriamine systems using near-infrared spectroscopy reveal that primary amines are nearly consumed before secondary amine conversion begins. Our simulations capture this early-stage behavior. The number of hydroxyl hydrogens on the CuO surface (H_{CuOH}) decreases slightly, suggesting a limited reaction at the interface.

To further characterize the spatial distribution of reaction events, we evaluated the depth-resolved reaction density $\rho_{\text{rxn}}(z)$:

$$\rho_{\text{rxn}}(z) = \frac{N_{\text{rxn}}(z)}{A\Delta z N_{\text{frames}}}$$

where $N_{\text{rxn}}(z)$ is the number of reactions observed in bin z , A is the cross-sectional area, and Δz is the bin width, set to 8 Å. Figure 5b shows $\rho_{\text{rxn}}(z)$ for epoxy–primary amine, epoxy–secondary amine, and epoxy–surface reactions, together with the atomic number density profiles $\rho(z)$ of epoxy, amine, and substrate. Interestingly, the reaction density near the interface is lower than that in the bulk, even though the local atomic densities of both epoxy and amine are higher near the interface. This finding is consistent with prior studies reporting densification near solid substrates, where restricted segmental mobility leads to reduced reaction rates, despite high local concentrations. In other words, our simulations confirm that interfacial reactions are suppressed in the near-surface region, in agreement with previous experimental and theoretical reports of reduced reactivity at polymer/solid interfaces.⁵⁰

A representative snapshot is shown in Figure 5c. Direct visualization confirms that epoxy ring-opening reactions occur at the CuO–OH surface, leading to covalent bond formation between the epoxy network and substrate. Even though the number of such interfacial bonds is small, their formation connects the polymer network to the substrate and is expected to significantly enhance adhesion, consistent with experimental observations.

5. CONCLUSION

The proposed framework successfully captures relative reactivity trends and qualitative mechanistic features. However, several limitations remain to be addressed. First, applying a bias potential disrupts the correspondence between simulation and physical time, preventing direct reproduction of the absolute rate constants. Second, although propagation events are systematically accelerated, the current implementation excludes other relevant pathways such as chain transfer and termination, which are essential for describing molecular-weight distributions in radical polymerization. Third, the accuracy of predicted barriers and kinetics is ultimately bound by the fidelity of the underlying uMLIP, which, in our case, does not fully resolve subtle energetic differences across monomers.

Thus, the following directions are recommended for future work: (i) development of higher-fidelity uMLIPs trained on more diverse and accurate reference data, including reaction barriers and transition-state information, to improve quantitative reliability; (ii) extension of the reaction catalog to incorporate side reactions and termination events, enabling a more comprehensive description of experimentally observed polymerization kinetics; and (iii) exploration of hybrid approaches that combine acceleration with approximate time rescaling or kinetic reconstruction, thereby bridging the gap

between qualitative trend-capture and quantitative rate prediction. In addition, coupling the present scheme with enhanced conformational sampling (e.g., replica exchange or basin-hopping strategies) may further improve the robustness of the predicted mechanistic pathways.

Overall, the TDBB approach, combined with a transferable uMLIP, provides a practical and versatile platform for simulating polymerization and curing processes. Its demonstrated ability to reproduce key features across diverse systems without system-specific parametrization highlights its promise as a foundation for the development of more quantitatively predictive and chemically comprehensive methodologies.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.5c08192>.

Additional figures, reaction pathway analyses, and simulation details (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Hodaka Mori – Preferred Networks, Inc., Tokyo 100-0004, Japan; orcid.org/0009-0006-2647-576X;
Email: hodakamori@preferred.jp

Authors

Shunsuke Tonogai – Preferred Networks, Inc., Tokyo 100-0004, Japan

Yu Miyazaki – Preferred Networks, Inc., Tokyo 100-0004, Japan

Akihide Hayashi – Preferred Networks, Inc., Tokyo 100-0004, Japan; orcid.org/0000-0001-9270-3050

Masayoshi Takayanagi – Data Science and AI Innovation Research Promotion Center, Shiga University, Hikone 522-8522, Japan; orcid.org/0000-0002-2458-0180

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acs.jpcb.5c08192>

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We express our sincere gratitude to So Takamoto of Preferred Networks, Inc.; Ryosuke Kuwabara of MITSUI & CO., LTD.; Sannohe Kunio of IBLC Co., Ltd.; and all their colleagues, for their valuable comments, insightful discussions, and continuous support.

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